Advanced Analytics Using Statistics: Comprehensive Cheatsheet

For CCEE (CDAC) Exam Preparation

Session 1 & 2: Introduction to Analytics & Data Analytics Lifecycle

Three Types of Analytics

- **Descriptive Analytics**: "What happened?" – Summarizes past data using counts, averages, dashboards[web:31]
- **Predictive Analytics**: "What is likely to happen?" – Uses historical data and models to forecast future outcomes[web:31]
- **Prescriptive Analytics**: "What should we do?" – Recommends actions based on predictions and constraints[web:31]

Data Analytics Lifecycle Phases[web:25][web:115][web:117]

Phase	Description	Key Activities
Discovery	Define business problem and feasibility	Problem definition, stakeholder identification, KPI setting, data source assessment[web:115]
Data Preparation	Clean and organize raw data	Handle missing values, remove duplicates, normalize, feature engineering, train/test split[web:115]
Model Planning	Select algorithms and evaluation metrics	Choose technique (regression/classification/clustering), define evaluation criteria, design experiments[web:25]
Model Building	Train and tune models	Hyperparameter tuning, feature selection, cross-validation, model comparison[web:117]
Implementation	Deploy model to production	Integration with systems, API setup, dashboards, real-time scoring[web:115]
Quality Assurance	Validate performance and stability	Model validation, performance monitoring, stress testing, A/B testing[web:117]
Documentation	Record all procedures and results	Data sources, preprocessing steps, model choice, performance metrics, audit trail[web:25]
Management Approval	Get formal sign-off	Present ROI in business terms, discuss risks and compliance[web:25]
Installation & Operation	Set up production systems	UAT, scheduled jobs, ongoing monitoring and maintenance, retraining as needed[web:115]

Intelligent Data Analysis (IDA)

Uses advanced ML/AI methods to **automatically** uncover patterns, make predictions, and support decisions in complex datasets.[web:30] Features include:

- Automated pattern recognition and anomaly detection
- Forecasting and predictions
- Decision support and optimization

Session 3 & 4: Probability Foundations

Sample Spaces and Events[web:49][web:62]

- **Sample Space (Ω)**: Set of **all possible outcomes** of a random experiment
  - Coin toss: Ω = {H, T}
  - Two coin tosses: Ω = {HH, HT, TH, TT}
- **Event**: Any subset of the sample space
  - Example: "At least one head" = {HH, HT, TH}
- **Probability** (equally likely outcomes): $P(A) = \frac{n(A)}{n(S)}$

Joint, Marginal, and Conditional Probability[web:54][web:60]

Type	Definition	Intuition
Joint $P(A \cap B)$	Probability A and B occur together	**Cell** in a cross-tab table
Marginal $P(A)$	Probability of A ignoring other variables, $P(A) = \sum_b P(A, b)$	**Row or column total** in a table
Conditional $P(A\|B) = \frac{P(A \cap B)}{P(B)}$	Probability of A **given** B occurred	**Zoom in** on cases where B happened

Key Relationship: $P(A \cap B) = P(A|B)P(B) = P(B|A)P(A)$

Bayes' Theorem[web:55][web:61]

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

Or for multiple causes:

$$P(A_i|B) = \frac{P(B|A_i)P(A_i)}{\sum_j P(B|A_j)P(A_j)}$$

Components:

- $P(A)$: **Prior** – belief about A before seeing B
- $P(B|A)$: **Likelihood** – probability B is true if A is true
- $P(A|B)$: **Posterior** – updated belief about A after seeing B

Real-world example: Disease testing where A = "has disease", B = "test positive"

- $P(A)$: disease prevalence (prior)
- $P(B|A)$: test sensitivity (likelihood)
- $P(A|B)$: probability of disease given positive test (what doctor cares about)

Session 5 & 6: Random Variables, Correlation & Covariance, Outliers

Random Variables[web:75][web:77]

- **Discrete**: Takes countable values (0, 1, 2, ...)
  - Examples: number of calls per hour, defects in batch
- **Continuous**: Takes any value in an interval
  - Examples: height, weight, time to failure

MCQ Rule: "Number of..." → Discrete; "Amount/time/weight/height/temperature" → Continuous

Covariance[web:83][web:89]

$$\text{Cov}(X,Y) = \frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})$$

- **Interpretation**:
  - Cov(X,Y) > 0: X and Y move **together** (positive association)
  - Cov(X,Y) < 0: X and Y move in **opposite** directions
  - Cov(X,Y) ≈ 0: No linear relationship
- **Limitation**: Not standardized; depends on units and scale

Correlation (Pearson's r)[web:83][web:86]

$$r_{XY} = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum(x_i - \bar{x})^2}\sqrt{\sum(y_i - \bar{y})^2}}$$

Range: −1 to +1 (dimensionless)

- r = +1: Perfect positive linear relationship
- r = −1: Perfect negative linear relationship
- r = 0: No linear correlation
- **Strength**: |r| ≥ 0.7 usually indicates strong correlation

Aspect	Covariance	Correlation
Range	−∞ to +∞	−1 to +1
Units	Depends on X, Y	Unitless
Comparability	Hard across datasets	Easy across variables

Outliers[web:81][web:87]

- **Definition**: Data points unusually far from most observations
- **Effects**: Can distort mean, SD, covariance, correlation; mislead models
- **Detection Methods**:
  - **Z-score**: $|z| > 3$ (>3 standard deviations from mean) → likely outlier
  - **IQR method**: Points beyond $Q1 - 1.5 \times IQR$ or $Q3 + 1.5 \times IQR$ → outliers

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Session 7 & 8: Probability Distributions

Discrete Distributions[web:104][web:109]

Distribution	Use Case	Parameters	Mean	Variance
Binomial	Fixed n trials, count successes	n, p	np	np(1-p)
Poisson	Events per fixed interval, no upper limit	λ	λ	λ
Geometric	Trials until first success	p	1/p	(1-p)/p²

Examples:

- Binomial: 20 products inspected, count defectives (fixed n=20)
- Poisson: Customers arriving per hour (no fixed max)

- Geometric: Coin tosses until first head

Continuous Distributions[web:104][web:111]

Distribution	Use Case	Key Properties
Uniform	Values equally likely in interval [a,b]	PDF = 1/(b-a), Mean = (a+b)/2
Exponential	Time between events in Poisson process	Mean = 1/λ, Memoryless property
Normal	Bell curve; symmetric around μ	68% within μ±σ, 95% within μ±2σ

Normal Distribution (Bell Curve):

- Mean = Median = Mode = μ
- Standard deviation = σ
- Applies to many natural measurements (heights, test scores, etc.)

Session 9 & 10: Descriptive Statistical Measures

Central Tendency[web:107]

Measure	Definition	Sensitivity to Outliers	Best For
Mean	Sum / n	**High** – most affected	Symmetric data
Median	Middle value when sorted	Low – robust	Skewed data, incomes
Mode	Most frequent value	None	Categorical data, peaks

MCQ Hints:

- "Affected most by extreme values?" → Mean
- "Best for skewed income data?" → Median
- "Most common category?" → Mode

Percentiles and Quartiles[web:104][web:107]

- **Percentiles**: Split data into 100 equal parts; pth percentile = value below which p% of data lie
- **Quartiles**: Split data into 4 equal parts
  - Q1 = 25th percentile
  - Q2 = 50th percentile = **Median**
  - Q3 = 75th percentile

Dispersion Measures

Measure	Definition	Units	Affected by Outliers
Range	Max − Min	Same as data	**Very much**
IQR	Q3 − Q1	Same as data	No – middle 50% only
Variance	$\frac{1}{n-1}\sum(x_i - \bar{x})^2$	**Squared**	Yes
Std Dev	$\sqrt{\text{Variance}}$	Same as data	Yes
CV	(SD / Mean) × 100%	**Unitless (%)**	Comparative

Coefficient of Variation (CV):

- Compares relative variability; scale-free
- Useful for comparing datasets with different means or units
- $\text{CV} = \frac{\sigma}{\mu} \times 100\%$

MCQ Hints:

- "Spread using only Q1 and Q3?" → IQR
- "Same units as data?" → SD, Range, IQR
- "Scale-free comparison?" → Coefficient of Variation

Quick Reference: 68-95-99.7 Rule (Normal Distribution)

- **68%** of data within μ ± 1σ
- **95%** of data within μ ± 2σ
- **99.7%** of data within μ ± 3σ

→ Z-score > 3 usually signals an outlier

Common MCQ Patterns & Answers

Pattern	Typical Answer
"Number of..."	Discrete RV, likely Binomial or Poisson
"Amount/time/weight"	Continuous RV
"Affects most by outliers"	Mean (not Median or Mode)
"Skewed data"	Median (not Mean)
"Counts with no fixed upper limit"	Poisson
"Counts with fixed n trials"	Binomial
"Time until event"	Exponential or Geometric
"Equal likelihood on interval"	Uniform
"Scale-free comparison"	Coefficient of Variation
"Middle 50% spread"	IQR
"Same units as data"	SD, Range, IQR (NOT Variance or CV)
"Posterior probability"	Bayes' theorem
"Zooming into one condition"	Conditional probability
"A and B together"	Joint probability

Study Tips for CCEE

1. **Lifecycle**: Know the **9 phases** in order; remember Discovery → Data Prep → Model Planning → Model Building → Implementation → QA → Documentation → Management Approval → Installation & Operation.
2. **Probability**: Master the distinction between **joint** (cells), **marginal** (totals), and **conditional** (zoom in).
3. **Distributions**: If "fixed n" → Binomial; if "counts per interval" → Poisson; if "continuous, symmetric" → Normal; if "waiting time" → Exponential.
4. **Descriptive Measures**: Remember Mean is sensitive to outliers; Median is robust. CV is for comparing datasets with different scales.
5. **Covariance vs Correlation**: Covariance is scale-dependent; correlation is always −1 to +1 and comparable.

6. **Outliers**: Check with Z-score ($|z| > 3$) or IQR rule before drawing conclusions about relationships.

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